Absolute Multi-Agent Performance Across Scale

Descriptive analysis from one to thirty-two agents

AlemDICE scaling study

July 2026

Abstract

This report analyzes absolute, rather than population-normalized, performance over two completed scaling campaigns. An earlier baseline screen contains three seeds at $N \in \{1, 2, 3, 4, 6\}$, while a matched comparison contains one 200-step episode for baseline and team-formation protocol at $N \in \{8, 16, 32\}$. The principal result is a positive protocol return advantage at every matched scale, accompanied by a larger share of coordinated achievement unlocks. The advantage is not uniform: acceleration is strongest from 8 to 16 agents and weakens from 16 to 32, exposing a concrete diminishing-yield regime. Because the matched cells contain only one seed, these are effect-size observations and mechanism diagnostics, not estimates of statistical significance.

1 Design and estimands

The two campaigns are deliberately analyzed separately. The earlier screen uses seeds 13100–13102 and reports means, observed ranges, and sample coefficients of variation. The matched campaign uses seed 14100 in every condition and reports paired descriptive contrasts. A dotted break in the all-scale figure marks the campaign boundary; no line is drawn between the six- and eight-agent baseline cells.

The main absolute estimands are

$$R_{\text{team}} = N\,\bar{R}_{\text{agent}}, \qquad A_{\text{sum}} = \sum_{i=1}^{N} \sum_{k} \mathbf{1}\{\text{agent } i \text{ first-unlocked achievement } k\}.$$

Thus R_{team} is the return summed over physical agents, and A_{sum} counts an achievement once for each agent that first unlocks it. Team-unique achievement types are reported separately and count each type at most once. These quantities answer whether adding agents produces more total work; they do not divide away population size.

N	Team return	Return/agent	Agent unlocks	Unique types	Return CV
1	7.1 [4.0, 11.0]	7.13	7.3 [4, 11]	7.3	49.9%
2	28.6 [19.0, 43.7]	14.28	16.0 [11, 22]	10.3	46.4%
3	65.0 [39.3, 88.0]	21.68	36.3 [30, 45]	19.3	37.6%
4	81.5 [60.2, 101.2]	20.37	42.3 [33, 48]	19.3	25.2%
6	71.3 [23.8, 103.6]	11.88	42.3 [24, 58]	15.0	58.9%

Table 1: Earlier baseline screen. Team return and agent unlocks are means with observed three-seed ranges in brackets. CV is the sample coefficient of variation of absolute team return. Each cell requested 200 steps; one three-agent episode terminated naturally at step 191.

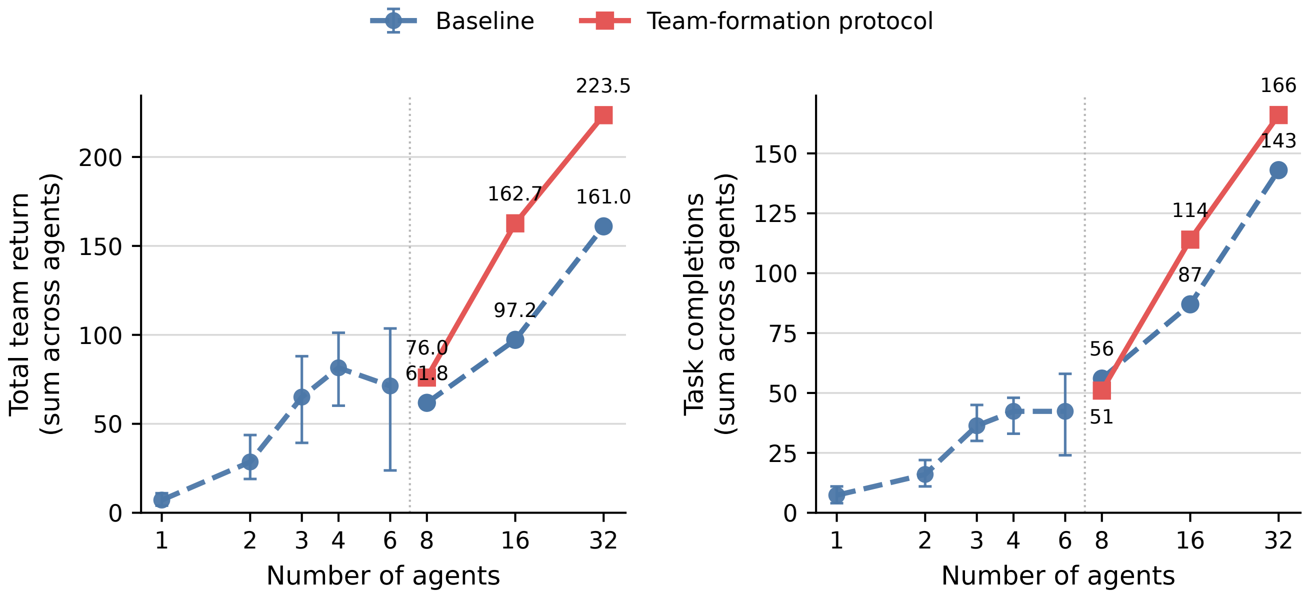

Figure 1: Absolute team output across the two campaigns. Task completions count only the first completion of each task by each agent. Whiskers at $N \leq 6$ are observed three-seed ranges. Points at $N \geq 8$ are single matched episodes. The vertical dotted line denotes the campaign boundary.

2 Earlier baseline screen: a capacity knee

Absolute baseline return rises sharply from 7.1 at one agent to 81.5 at four agents, an 11.4× increase for a 4× population increase. Summed achievement unlocks increase 5.8×, from 7.3 to 42.3. This superlinear early region is consistent with task parallelism and access to multi-agent opportunities, but it is not a clean causal estimate: population also changes spawn geometry, pressure, specialization balance, and available coordination.

The screen then exhibits a knee. Moving from four to six agents reduces mean absolute return by 12.5% (81.5 to 71.3), leaves mean summed unlocks exactly flat at 42.3, and reduces unique team achievement types from 19.3 to 15.0. The six-agent return CV is 58.9%, the largest in the screen, compared with 25.2% at four agents. This is evidence of unstable saturation in the baseline-as-implemented, not evidence that six agents are intrinsically worse: the original campaign documents a geometric-feasibility confound at $N = 6$.

N	Return B	Return P	Δ return	Unlocks B	Unlocks P	Δ unlocks	Unique B/P
8	61.8	76.0	+23.0%	56	51	-8.9%	14/19
16	97.2	162.7	+67.4%	87	114	+31.0%	15/16
32	161.0	223.5	+38.8%	143	166	+16.1%	18/21

Table 2: Matched absolute performance. B denotes baseline and P denotes the team-formation protocol. Each value is one 200-step episode at the same seed; deltas are descriptive, not confidence intervals.

3 Matched protocol comparison

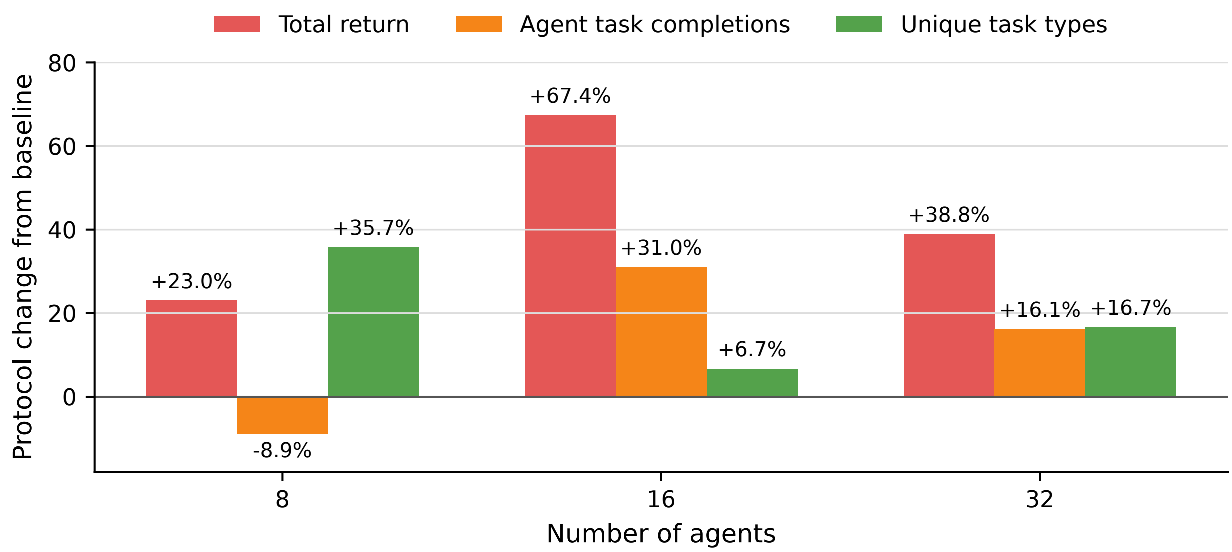

Figure 2: Protocol change relative to the matched baseline. Task completions count only the first completion of each task by each agent. Return improves at all three scales. The eight-agent cell combines higher return and broader team-unique coverage with fewer summed per-agent task completions, revealing a quality/breadth versus repetition tradeoff.

3.1 Finding 1: positive return signal at every matched scale

The protocol increases absolute return by 23.0% at eight agents, 67.4% at 16 agents, and 38.8% at 32 agents. In raw units the gains are 14.2, 65.5, and 62.5. The 32-agent result is therefore not merely an artifact of a percentage denominator: it retains nearly the full 16-agent absolute gain. Nevertheless, one seed cannot establish robustness or a population-level confidence interval.

3.2 Finding 2: the eight-agent result favors breadth over repetition

At eight agents the protocol produces 51 summed unlocks versus 56 for baseline (−8.9%), yet reaches 19 unique team achievement types versus 14 (+35.7%) and earns 23.0% more return. The protocol therefore did not win by having every agent repeat more easy milestones. It covered more distinct achievement types and shifted activity toward coordinated events. This is a useful mechanism signal: raw unlock volume alone would incorrectly classify the eight-agent result as worse.

N	Base B	Coord B	Share B	Base P	Coord P	Share P	Coord count ratio
8	54	2	3.6%	43	8	15.7%	4.00×
16	83	4	4.6%	103	11	9.6%	2.75×
32	138	5	3.5%	149	17	10.2%	3.40×

Table 3: Composition of summed agent first-unlocks. “Base” denotes individual-task achievements and “Coord” coordinated achievements.

4 Coordination composition

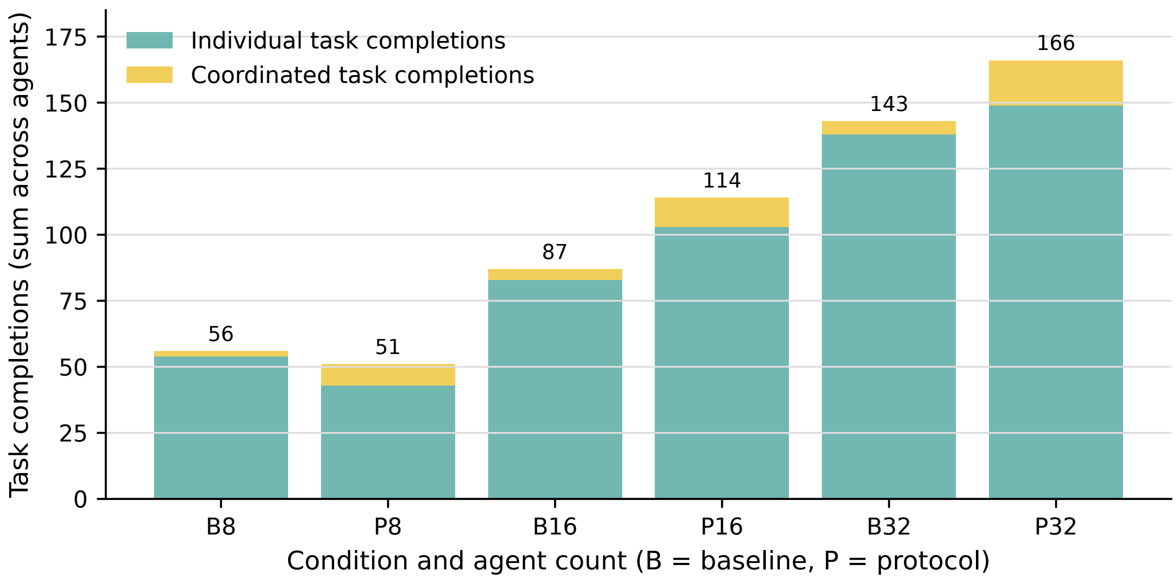

Figure 3: Individual and coordinated first task completions summed across agents. Each adjacent pair compares baseline (B) and protocol (P) at the same population.

4.1 Finding 3: coordinated achievements are the most consistent mechanism signal

The coordinated-unlock count rises from 2 to 8 at $N = 8$, from 4 to 11 at $N = 16$, and from 5 to 17 at $N = 32$: multipliers of 4.00×, 2.75×, and 3.40×. Coordinated unlocks constitute only 3.5–4.6% of baseline unlocks, but 9.6–15.7% under the protocol. Unlike total unlock volume, this directional signal is positive at every matched scale and directly reflects the intended behavioral mechanism.

The effect is not only substitution. At 16 and 32 agents, individual-task unlocks also increase (83 to 103 and 138 to 149), while coordinated unlocks increase more sharply. At eight agents individual-task unlocks fall from 54 to 43, making that cell the clearest example of specialization reducing redundant individual progress while expanding team-level breadth.

Condition	Interval	α_R	α_A	Marginal return	Marginal unlocks
Baseline	8→16	0.653	0.636	4.42	3.88
Baseline	16→32	0.728	0.717	3.99	3.50
Protocol	8→16	1.098	1.160	10.84	7.88
Protocol	16→32	0.458	0.542	3.80	3.25

Table 4: Local scaling diagnostics. Elasticity $\alpha = \log(Y_1/Y_0)/\log(N_1/N_0)$; $\alpha = 1$ is linear absolute scaling. Marginal columns report added output per added agent.

5 Scaling efficiency and diminishing marginal yield

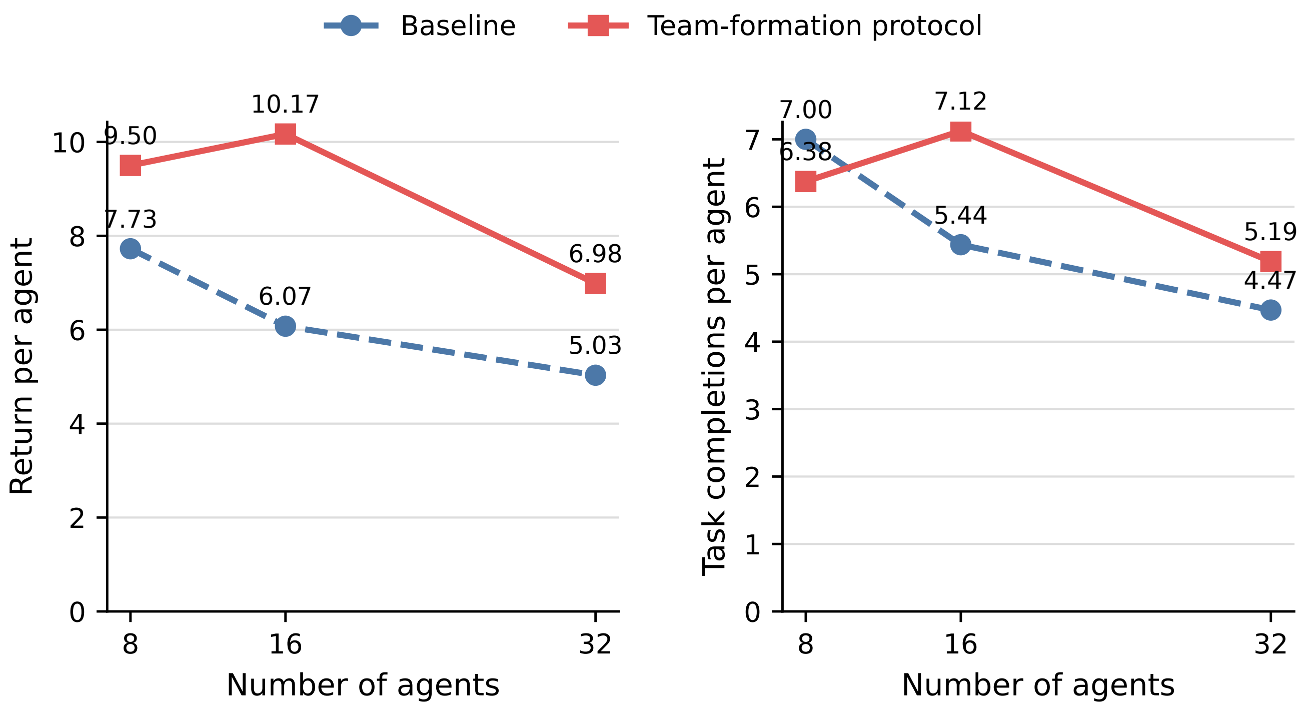

Figure 4: Per-agent efficiency in the matched campaign. These normalized diagnostics explain the absolute curves; they are not the primary performance estimands.

For output Y, define the local scaling elasticity $\alpha = \log(Y_1/Y_0)/\log(N_1/N_0)$. Linear absolute scaling has $\alpha = 1$, sublinear scaling has $\alpha < 1$, and superlinear scaling has $\alpha > 1$.

5.1 Finding 4: the protocol's strongest scaling phase is 8 to 16

From 8 to 16 agents, protocol return has elasticity 1.098 and unlock volume has elasticity 1.160, both slightly superlinear. Baseline elasticities are 0.653 and 0.636. Each added agent in this interval yields 10.84 additional return units and 7.88 additional unlocks under the protocol, versus 4.43 and 3.88 for baseline.

From 16 to 32 agents, protocol elasticities fall to 0.458 for return and 0.542 for unlocks. The protocol remains absolutely better, but its marginal yield falls to 3.80 return units and 3.25 unlocks per added agent, close to baseline's 3.99 and 3.50. This pattern localizes the next research problem: beyond 16 agents, additional team structure preserves an output lead but no longer creates superior marginal scaling.

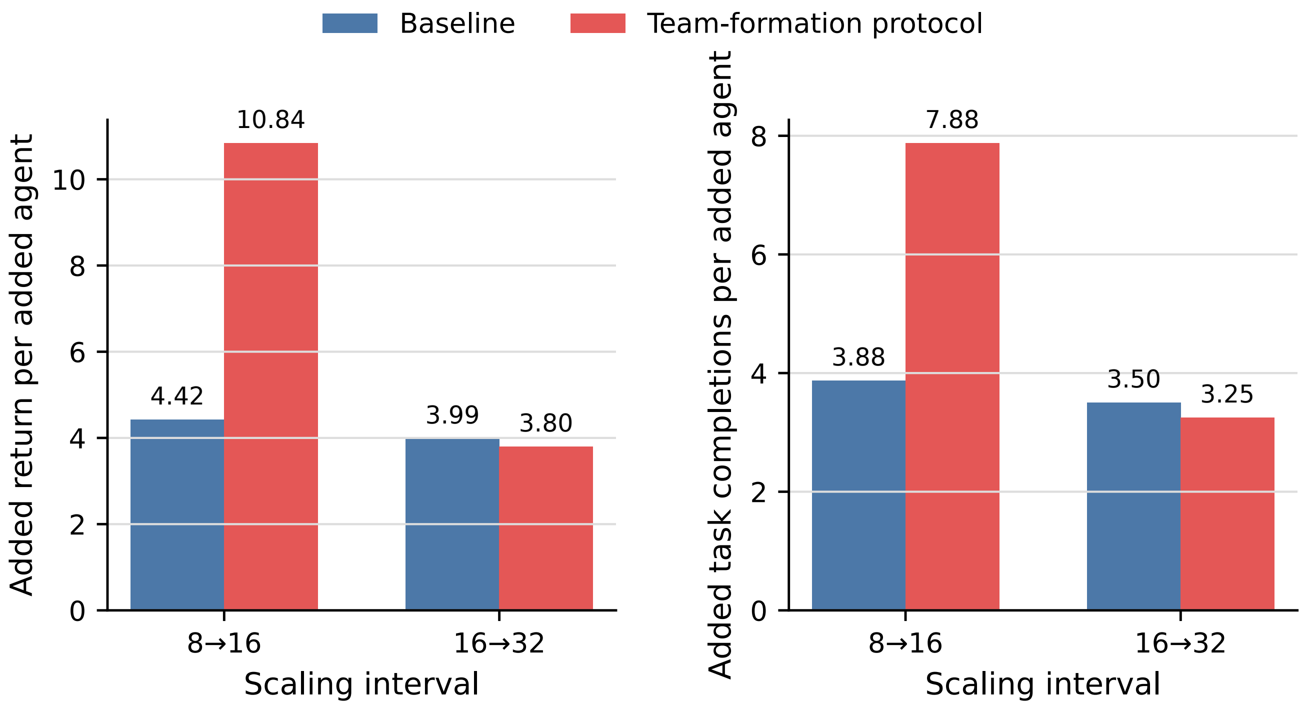

Figure 5: Marginal absolute output per added physical agent. Task completions count only the first completion of each task by each agent. The protocol's large advantage in the 8-to-16 transition nearly disappears in the 16-to-32 transition.

5.2 Finding 5: the protocol mitigates, but does not eliminate, per-agent decay

Across 8 to 32 agents, baseline return per agent falls from 7.73 to 5.03 (-34.9%); protocol return per agent falls from 9.50 to 6.98 (-26.5%). Summed unlocks per agent fall from 7.00 to 4.47 for baseline and from 6.38 to 5.19 for the protocol. The protocol therefore retains more useful output per agent at the largest scale, especially in achievements, but does not achieve scale-invariant efficiency.

6 Interpretation and next experiments

The data support three bounded claims. First, the protocol has a positive absolute-return signal at all matched scales. Second, it consistently increases coordinated achievement activity and team-level breadth. Third, the scaling mechanism changes regime: strong gains through 16 agents give way to diminishing marginal yield by 32 agents.

The following experiments are the shortest path from signal to evidence:

1. Repeat every $N \in \{8, 16, 32\}$ matched cell for at least three, preferably ten, common seeds. Report paired seed differences and bootstrap intervals rather than intervals over unmatched episodes.
2. Instrument formation overhead: messages, delivered bytes, inactive turns, reassignment counts, and time-to-first coordinated unlock. This can distinguish communication saturation from environmental saturation.
3. At 32 agents, compare one global team with two and four independent teams while holding the physical population and task cap fixed. The current marginal-yield collapse predicts that partitioning should restore local planning efficiency.
4. Preserve the achievement decomposition. Primary outcomes should include return, summed unlocks, unique types, and coordinated unlocks; any one of these alone misses the eight-agent tradeoff.

7 Limitations

The high-scale comparison has one seed per cell, so no claim here is a statistical significance claim. The low- and high-scale campaigns use different seeds and experimental configurations; their points are juxtaposed to show coverage, not treated as one matched response curve. Achievement first-unlocks are cumulative binary events, not counts of repeated task completion. Finally, absolute output naturally tends to rise with population; the efficiency and marginal-yield diagnostics are required to determine whether those gains are proportional to the additional agents.